

# High-Performance PE-BN/PVDF-HFP Bilayer Separator for Lithium-Ion Batteries

Muhammad Waqas, Shamshad Ali, Weiqiang Lv, Dongjiang Chen, Bismark Boateng, and Weidong He\*

Highly efficient and thermally stable polyethylene-hexagonal boron nitride/poly(vinylidene fluoride-hexafluoropropylene) (PE-BN/PVDF-HFP) bilayer separator with microporous structure is prepared, for the first time with a wet chemistry method. The incorporation of hexagonal boron nitride particles in PE matrix promotes the interfacial interaction between PE and PVDF-HFP layers to prevent separation of layers and suppresses dendrite growth owing to the strong adsorption energy with polymers, large interactive surface area, and superior mechanical capability. The PVDF-HFP layer provides additional thermally stable backbone while its inherent hydrophilic property and highly porous structure improve the overall performance of the separator. The bilayer separator owns a high electrolyte uptake of 348% and a thermal shrinkage of 6.6% upon annealing at 140 °C for 1 h. The lithium iron phosphate/lithium cell with the as-prepared separator owns a high ionic conductivity up to  $4.4 \times 10^{-4} \text{ S cm}^{-1}$ , and a room-temperature capacity of 120 mAh g<sup>-1</sup> at 2 C with a 95% capacity retention after 500 cycles and a capability of 108 mAh g<sup>-1</sup> at 4 C. The PE-BN/PVDF-HFP separator is a promising alternative of traditional multilayer separators for high-performance lithium-ion batteries.

## 1. Introduction

Separator is one of the essential components of lithium-ion batteries (LIBs), which is used to prevent physical contact between anode and cathode to avoid internal short circuits, preserve liquid electrolyte, and permitting the rapid migration of lithium ions during cycling process.<sup>[1–4]</sup> The most commonly used separators in LIBs are polyolefin separators predominantly polypropylene (PP), polyethylene (PE), and their multilayer formations like PE/PP or PP/PE/PP owing to their high tensile strength and shutdown

ability. Nevertheless, the polyolefin separators suffer from severe dimensional instability at elevated temperatures and poor compatibility with liquid electrolytes due to hydrophobic surface character, and have poor capability to retain electrolyte.<sup>[5–10]</sup> Conventionally, polyolefin separators are prepared through dry or wet processes and their tensile strength, porosity, or Gurley number vary with respect to methods of preparation.<sup>[11,12]</sup> However, these conventional methods are very complex and require high-temperature treatment.<sup>[13–15]</sup> During the past decades, intensive efforts have been made to improve the thermal stability, wettability, and electrolyte retention of existing bilayer or trilayer polyolefin separators. Among those efforts, surface coating,<sup>[16–18]</sup> surface grafting,<sup>[19–21]</sup> and blending<sup>[22]</sup> with various inorganic materials including silica (SiO<sub>2</sub>), alumina (Al<sub>2</sub>O<sub>3</sub>), and titania (TiO<sub>2</sub>), and polymers were mostly focused. Surface coating method has drawn great attention

among these approaches on account of the facile process to enhance the performance of polyolefin separators. Polyvinylidene fluoride (PVDF), poly(vinylidene fluoride-hexafluoropropylene) (PVDF-HFP), and their composites have been extensively studied as surface coating of polyolefin separators to overcome the aforementioned drawbacks.<sup>[23–28]</sup> But the major drawback of surface coating method is the separation of the coated layer during charge–discharge process due to the low surface energy of polyolefin matrix, results in the sharp rise in impedance of separator which further deteriorates the cell performance.<sup>[6]</sup>

The incorporation of electrolyte-philic and thermally stable additives in PE matrix having strong interfacial interaction with polymers matrices is the lasting solution to problems discussed above. 2D materials have great research interests in many fields owing to their ultrathin structural features and chemical functionalities. 2D hexagonal boron nitrides (*h*-BN) have high mechanical strength, high thermal stability, and high chemical stability.<sup>[29–31]</sup> In *h*-BN, B and N bonds own partial ionic characteristics owing to the high electronegativity of N atoms with sp<sup>2</sup>-hybridized B–N  $\sigma$  bonds. Moreover, in the N atom p<sub>z</sub> orbital, the lone pair electrons are partially delocalized with empty B p<sub>z</sub> orbital. These unique characteristics lead to the superior thermal, electrochemical, and electrically insulating properties of *h*-BN.<sup>[32–34]</sup> The highly interactive surface area and strong adsorption energy of *h*-BN enable interfacial bonding and functionalities of bonded materials at high temperatures.<sup>[32]</sup>

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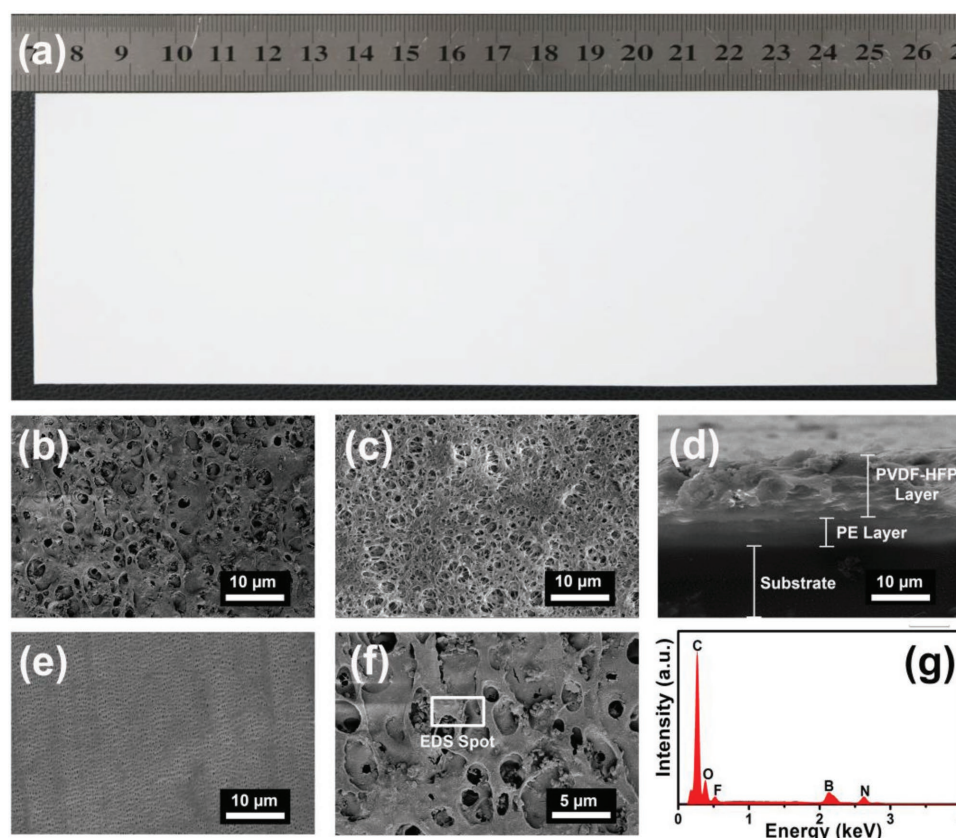
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In this work, PE-BN/PVDF-HFP bilayer separator has been prepared, for the first time with a wet chemistry method, which is more convenient, efficient, and cost effective in contrast to conventional dry and wet methods. The incorporation of *h*-BN in PE matrix improves the thermal stability, suppresses dendrite growth, and promotes the strong bonding between the PE and the PVDF-HFP layers attribute to the superior mechanical strength, large interactive surface area, and strong adsorption energy. The PVDF-HFP coating on the polyethylene-hexagonal boron nitride (PE-BN) layer maintains the thermal stability beyond the melting temperature of PE to avoid internal shorting at elevated temperature while its inherent hydrophilic property and highly porous structure improve the performance of the cell. The high cycle performance and C-rate capabilities of LFP/Li cell with PE-BN/PVDF-HFP separator are achieved at room temperature and elevated temperatures, which is attributed to high electrolyte uptake, high ionic conductivity, and strong interaction between PE-BN and PVDF-HFP layers. The advantages of as-prepared bilayer separator make it promising substitute to commercial bilayer or trilayer polyolefin separators.

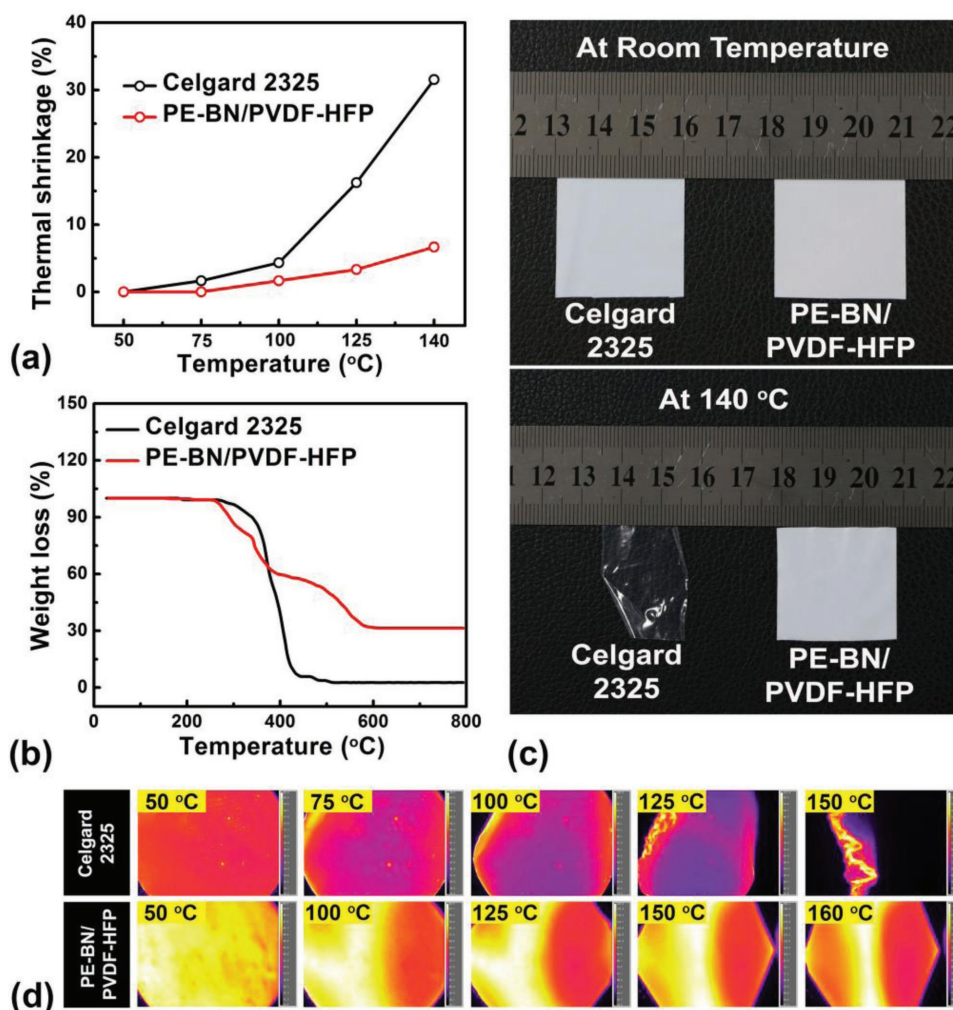
## 2. Results and Discussion

Figure 1a shows the photograph of uniform PE-BN/PVDF-HFP bilayer separator with 40  $\mu\text{m}$  thickness prepared through wet

chemistry method. Figure 1b–d depicts the scanning electron microscopic (SEM) images of the surface (top and bottom) and cross-sectional morphologies of PE-BN/PVDF-HFP bilayer separator. The top and bottom surfaces of as-prepared separator contain microporous structure with several long micropores which are attributed to preparation methodology and evaporation of different solvents during heating under vacuum. Moreover, the cross-section image of the as-prepared separator confirms the bilayer structure of the PE-BN/PVDF-HFP separator. Figure 1e shows the surface morphology of commercial trilayer separator (Celgard 2325) with long elliptical pores formed through stretching technology. The porosity, mean pore diameter, and pore size distribution of Celgard 2325 and PE-BN/PVDF-HFP separators are analyzed by using mercury intrusion technique, as shown in Figure S1 (Supporting Information). The PE-BN/PVDF-HFP bilayer separator owns a high porosity of 50.8% with a mean pore diameter of 0.35  $\mu\text{m}$  and Celgard 2325 owns a porosity of 28.7%, as shown in Table S1 (Supporting Information). Owing to the interconnected microporous structure, large pore size, and high porosity, the bilayer separator absorbs and retains more electrolyte, which improves the cyclic performance and rate capability. The high porosity and interconnected microporous structure of the as-prepared bilayer separator are attributed to the preparation of separator through the wet chemistry method using two different solvents with different evaporation rates.<sup>[9,35,36]</sup> The distribution of different elements in PE-BN/PVDF-HFP bilayer separator is



**Figure 1.** a) Photograph of PE-BN/PVDF-HFP bilayer separator. SEM images of PE-BN/PVDF-HFP separator: b) bottom surface, c) top surface, d) cross-section view, e) top surface SEM image of Celgard 2325 separator at a scale of 10  $\mu\text{m}$ , and f,g) SEM and EDS images of PE-BN/PVDF-HFP separator confirming the presence of different elements in the 40  $\mu\text{m}$  thick separator.



**Figure 2.** a) Thermal shrinkage graph of Celgard 2325 and PE-BN/PVDF-HFP separators at different temperatures for 1 h, b) TGA curves of Celgard 2325 and PE-BN/PVDF-HFP separators, c) thermal shrinkage photographs of the Celgard 2325 and PE-BN/PVDF-HFP separators before and after annealing at 140 °C for 1 h, and d) FLIR images of Celgard 2325 and PE-BN/PVDF-HFP separators showing distribution of heat with 40  $\mu$ m thickness.

observed by energy-dispersive X-ray spectroscopy (EDS), as shown in Figure 1f,g. The EDS graph confirms the incorporation of *h*-BN particles and PVDF-HFP polymer coating from the presence of uniformly distributed B, N, F, C, and O elements in PE-BN/PVDF-HFP bilayer separator.

Thermal and mechanical stability of a separator are essential factors to evaluate the safety characteristics of LIBs. In order to investigate the dimensional stability, the thermal shrinkage of separators is observed at different temperatures (50–140 °C) for 1 h as shown in Figure 2a. The trilayer Celgard 2325 separator exhibits poor dimensional stability at 140 °C for 1 h with 32% thermal shrinkage in contrast to bilayer PE-BN/PVDF-HFP separator which owns 6.6%. The photographs of dimensional changes in Celgard 2325 and PE-BN/PVDF-HFP separators before and after annealing at 140 °C for 1 h with 40  $\mu$ m thickness are given in Figure 2c. The SEM images of Celgard 2325 and PE-BN/PVDF-HFP separators after annealing at 140 °C for 1 h are shown in Figure S2 (Supporting Information). After annealing at 140 °C, the trilayer (PP/PE/PP) Celgard 2325 separator shrinks severely due to the melting of intermediate

layer (PE layer) and blocks the pores of PP layer. The size of pores becomes smaller with decreasing porosity and the shape of pores turns to round from elliptical porous structure. However, the PE-BN/PVDF-HFP separator maintains similar interconnected porous structure in both top and bottom surfaces. The similar porous structure of the as-prepared separator after annealing illustrates the enhanced thermal and dimensional stability of the bilayer separator, which is owing to the incorporation of *h*-BN particles in PE matrix and the coating of the PVDF-HFP layer. The forward looking infrared (FLIR) images of Celgard 2325 and PE-BN/PVDF-HFP separators are shown in Figure 2d. The infrared thermography images represent thermal variations on the surface of separators with rise in temperature (50–160 °C) with respect to time. The PE-BN/PVDF-HFP bilayer separator maintains its integration at 160 °C after 300 s whereas Celgard 2325 (trilayer PP/PE/PP separator) starts to deform at 100 °C and completely melts at 150 °C after 300 s.

The thermogravimetry (TG) curves of Celgard 2325 and PE-BN/PVDF-HFP separators are shown in Figure 2c. The Celgard 2325 separator starts to degrade from 260 °C and

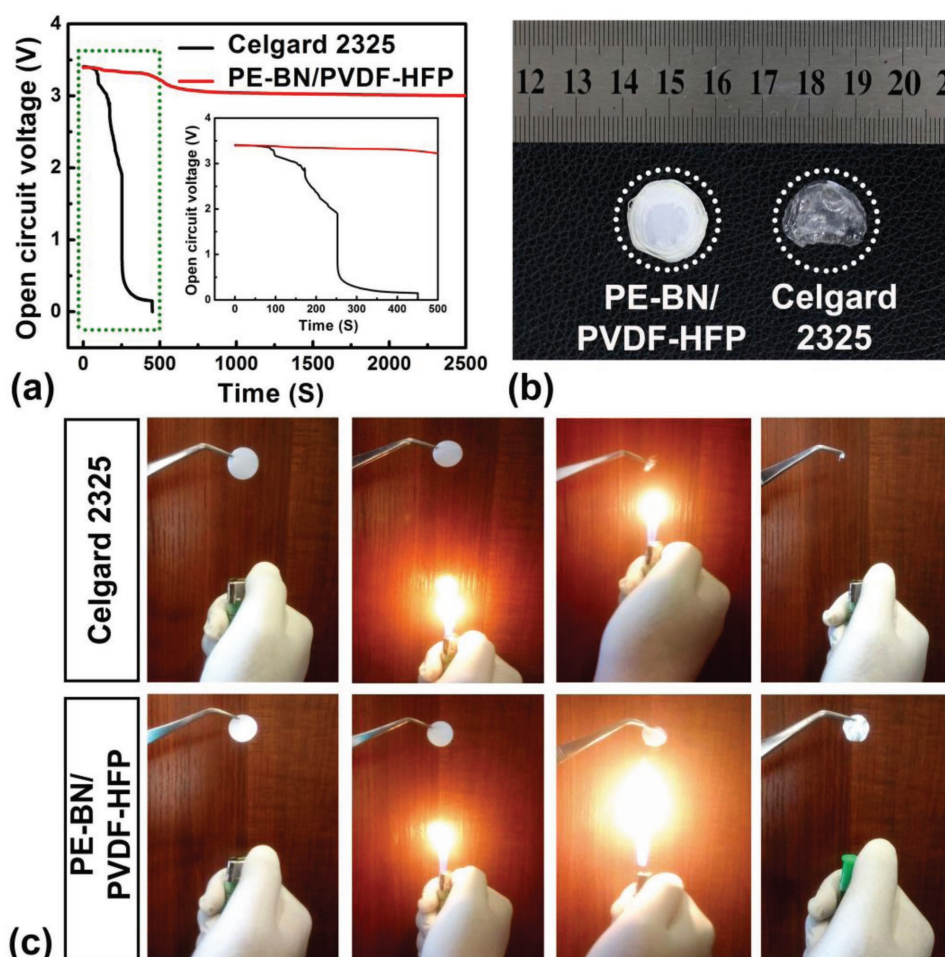


completely deformed with a weight loss of 100% at 425 °C. However, the PE-BN/PVDF-HFP bilayer separator represents three-stage TG graph, owing to the presence of three different materials (PE, BN, and PVDF-HFP). The overall weight loss of 75% is observed from bilayer separator between 260 and 680 °C. This enhancement in thermal stability of PE-BN/PVDF-HFP separator is attributed to the incorporation of thermally stable *h*-BN particles with PE matrix, coating of thin PVDF-HFP layer,<sup>[23,37]</sup> and method of preparation without involvement of stretching process.<sup>[14,38]</sup>

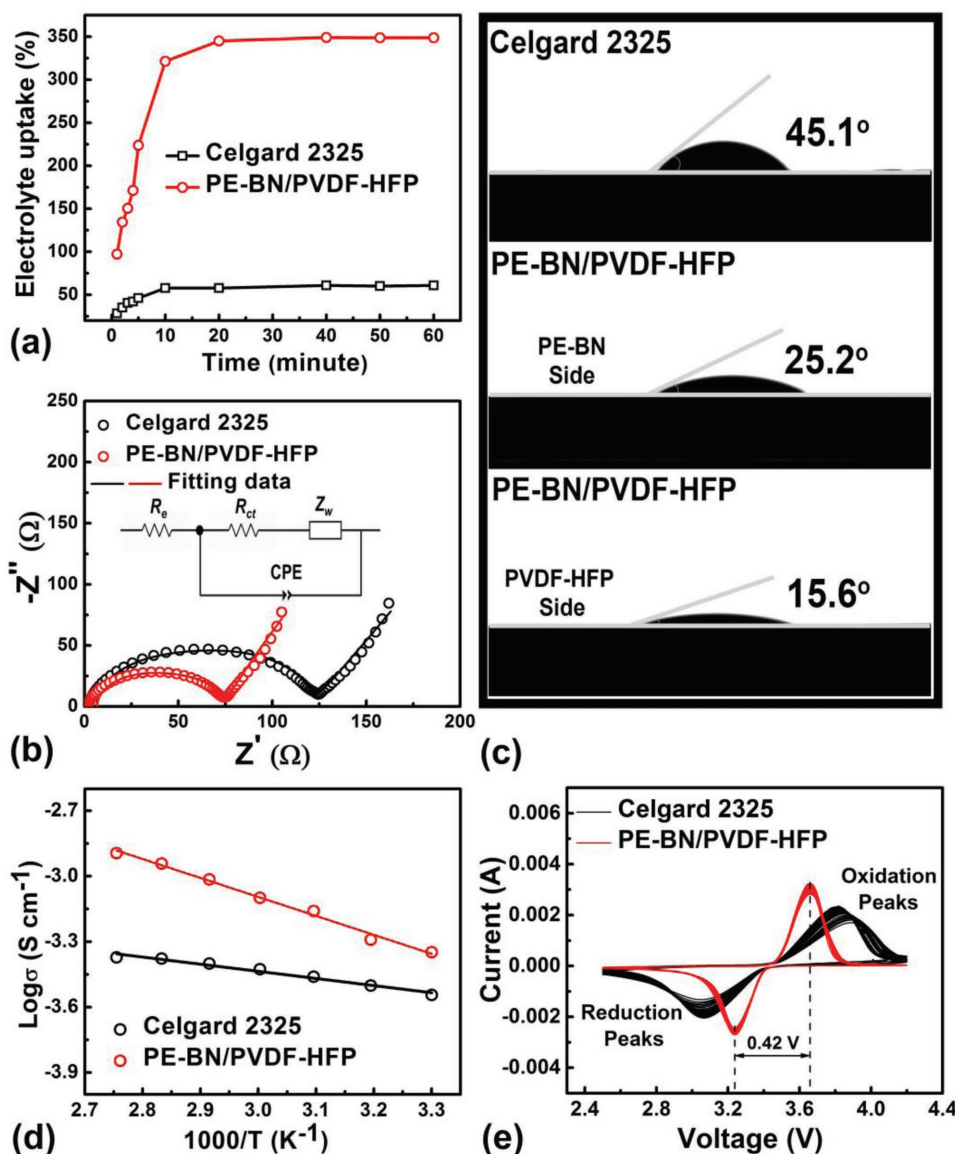
The battery separator must have enough strength to withstand during assembly procedure of batteries and prevent internal short circuits caused by lithium dendrites growth. Figure S3 (Supporting Information) shows the strain–stress curves of Celgard 2325 and PE-BN/PVDF-HFP separators. The tensile strength and the elongation rate of PE-BN/PVDF-HFP separator in *x* and *y* directions (22.01 MPa, 67.9%) are higher as compared to the Celgard 2325 separator (11.5 MPa, 31.3%) in the transverse direction (TD) due to the strong bonding between PE-BN and PVDF-HFP layers and bilayer structure of the as-prepared separator.<sup>[39]</sup> The observed tensile strength (22.01 MPa) of PE-BN/PVDF-HFP bilayer separator

withstands the force experienced in the assembly process of LIBs.<sup>[14]</sup>

The safety performance of LFP/Li cells is further investigated with open-circuit voltage (OCV) measurements. **Figure 3a** shows the OCV curves of LFP/Li cells assembled with Celgard 2325 and PE-BN/PVDF-HFP separators. The OCV of LFP/Li cell with Celgard 2325 drops suddenly to 0 V after 450 s at 150 °C, owing to the severe thermal shrinkage of separator and results in short circuit inside the cell. The cell with PE-BN/PVDF-HFP separator did not exhibit the sharp drop in voltage during the entire heating process due to high thermal stability with robust microstructure and bilayer formation, however, the slow voltage drop is observed due to the self-discharge at elevated temperatures.<sup>[18]</sup> The photographic images of Celgard 2325 and PE-BN/PVDF-HFP separators after OCV measurements are given in **Figure 3b**. The Celgard 2325 separator shows serious shrinkage and binds to the cathode in contrast to PE-BN/PVDF-HFP separator which maintains its dimensional stability during the entire measurement process. The fire combustion behavior of Celgard 2325 and PE-BN/PVDF-HFP separators is given in **Figure 3c**. When the commercial trilayer separator (Celgard 2325) set on fire, it catches the fire



**Figure 3.** a) OCV measurement of LFP/Li cells based on Celgard 2325 and PE-BN/PVDF-HFP separators, b) photographs of Celgard 2325 and PE-BN/PVDF-HFP separators after OCV measurement at 150 °C, and c) the combustion test photographs of Celgard 2325 and PE-BN/PVDF-HFP separators with 40 μm thickness.



**Figure 4.** a) Electrolyte uptake behavior of Celgard 2325 and PE-BN/PVDF-HFP separators at room temperature for different time intervals, b) EIS spectra of Celgard 2325 and PE-BN/PVDF-HFP separators, c) contact angle measurement images of Celgard 2325 and PE-BN/PVDF-HFP separators with liquid electrolyte, d) temperature-dependent ionic conductivities, and e) CV graphs of LFP/Li cells with Celgard 2325 and PE-BN/PVDF-HFP separators with 40  $\mu\text{m}$  thickness.

immediately and shows worst shrinkage. However, the PE-BN/PVDF-HFP bilayer separator did not burn when put on fire and maintains the dimensional stability. The as-prepared separator exhibits remarkable flame retarding ability, which is attributed to the oxidation-resistant capability of incorporated *h*-BN particles in PE matrix.<sup>[40–43]</sup> The robust thermal stability and flame retarding ability of the PE-BN/PVDF-HFP separator ensure the safety of LIBs in high-temperature environment.

Figure 4a shows the electrolyte uptake profiles of Celgard 2325 and PE-BN/PVDF-HFP separators as a function of soaking time. These separators are immersed in liquid electrolyte for different time intervals inside the glove box to analyze the electrolyte uptake. As compared to Celgard 2325 separator, the PE-BN/PVDF-HFP separator soaks up electrolyte so rapidly.

After 60 min of immersion in liquid electrolyte, the uptake of PE-BN/PVDF-HFP and Celgard 2325 separators is 348 and 60.8%, respectively. The significant enhancement in electrolyte uptake of PE-BN/PVDF-HFP separator is attributed to interconnected microporous structure, high porosity, incorporated *h*-BN particles in PE matrix, and an additional layer of PVDF-HFP. However, the electrolyte retention of PE-BN/PVDF-HFP separator is also higher than the Celgard 2325 separator, as shown in Figure S4 (Supporting Information). The contact angle measurements are conducted with liquid electrolyte to further investigate the effect of *h*-BN particles in PE matrix and coating of PVDFHFP layer on surface properties of the bilayer separator. As shown in Figure 4c the contact angle of PE-BN/PVDF-HFP separator is lower than the Celgard 2325 separator, which

confirms the electrolyte-philic nature of as-prepared separator. Even from PE-BN side (bottom surface) of PE-BN/PVDF-HFP separator, the contact angle is still lower than Celgard 2325, due to the polar nature of incorporated *h*-BN particles. Thus, the high electrolyte wettability of bilayer separator is beneficial to enhance the cycle performance and C-rate capabilities.

The electrochemical impedance spectra (EIS) profiles of LFP/Li cells based on Celgard 2325 and PE-BN/PVDF-HFP separators at room temperature are shown in Figure 4b. The EIS profile consists of an inclined line in low-frequency region corresponding to diffusion of lithium ions in electrode active material, semicircle in medium frequency region denotes charge transfer resistance accompanied with lithium-ion migration between electrodes and electrolyte interface. The intercept with  $\omega$ -axis is in high-frequency region and corresponding to ohmic resistance associated with ionic conductivity of the cell.<sup>[3,44]</sup> The electrical equivalent circuit for EIS profile is also prepared to further describe the impedance spectra (inset in Figure 4b). The circuit parameters  $Z_o$ ,  $R_{ct}$ ,  $R_e$ , and CPE of equivalent circuit represent Warburg impedance, charge transfer resistance, ohmic resistance, and constant phase element, respectively. The fitting impedance curves also agree with the experimental EIS data obtained by using the equivalent circuit. The EIS equivalent circuit results are shown in Table S1 (Supporting Information). The fitting data show that PE-BN/PVDF-HFP separator has a lower  $R_{ct}$  value (67.7  $\Omega$ ) as compared to Celgard 2325 (118.1  $\Omega$ ) separators. The lower  $R_{ct}$  value of PE-BN/PVDF-HFP separator confirms the excellent interface stability and high electrolyte retention of PE-BN/PVDF-HFP.<sup>[45]</sup>

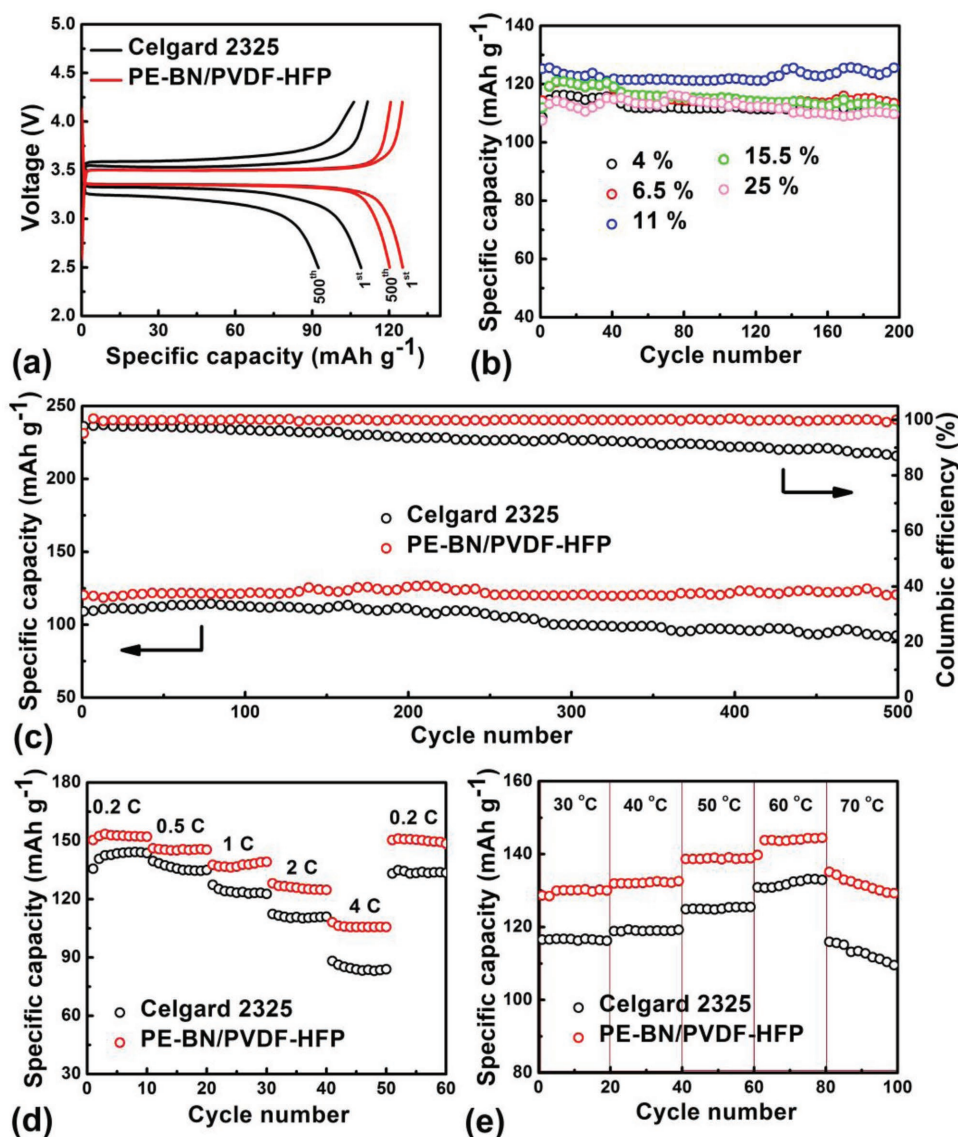
The AC impedances of block cells based on electrolyte soaked PE-BN/PVDF-HFP and Celgard 2325 separators are observed in Figure S5 (Supporting Information). The temperature dependence ionic conductivities of aforementioned separators are measured from these AC impedance plots by using Equation (S1) (Supporting Information). The ionic conductivities of PE-BN/PVDF-HFP separator are  $4.4 \times 10^{-4}$  S cm<sup>-1</sup> at 30 °C and  $1.2 \times 10^{-3}$  S cm<sup>-1</sup> at 90 °C while Celgard 2325 owns  $2.8 \times 10^{-4}$  S cm<sup>-1</sup> at 30 °C and  $4.2 \times 10^{-4}$  S cm<sup>-1</sup> at 90 °C as shown in Figure 4d. The ionic conductivities increase gradually with temperature and exhibit Arrhenius behavior. The higher ionic conductivity of bilayer PE-BN/PVDF-HFP separator is due to the high electrolyte uptake and uniform interconnected pore distribution, which confirm the transportation of lithium ions more efficiently. The cyclic voltammetry (CV) profiles of LFP/Li cells based on Celgard 2325 and PE-BN/PVDF-HFP separators measured at a scanning rate of 0.5 mV s<sup>-1</sup> between 2.5 and 4.2 V at room temperature for 50 cycles are shown in Figure 4e. The CV curves of LFP/Li cells with both separators exhibit symmetric oxidation and reduction peaks during extraction and insertion of lithium ions, respectively. The oxidation and reduction peaks of LFP/Li cell based on PE-BN/PVDF-HFP separator appear at 3.65 and 3.24 V, during anodic and cathodic scans, respectively. The PE-BN/PVDF-HFP separator owns higher oxidation-reduction peaks compared to LFP/Li cell assembled with Celgard 2325 separator. This reveals the enhancement in reversibility of lithium insertion and extraction reaction in LFP/Li cell with PE-BN/PVDF-HFP separator. The 1st, 10th, 20th, 40th, and 50th CV cycle profiles of LFP/Li cells with PE-BN/PVDF-HFP and Celgard 2325 separators are

shown in Figure S6 (Supporting Information). The stable and long CV cycles performance of LFP/Li cell with PE-BN/PVDF-HFP bilayer separator confirms the improved compatibility of the as-prepared separator with electrodes compared with Celgard 2325.

The electrochemical performance of LFP/Li cells based on Celgard 2325 and PE-BN/PVDF-HFP separators is evaluated in terms of cyclic performance and C-rate capability at room temperature and elevated temperatures. Figure 5a depicts the 1st and 500th charge-discharge profiles of LFP/Li cells based on Celgard 2325 and PE-BN/PVDF-HFP separators at 2 C with an operating voltage ranging from 2.5 to 4.2 V. The specific discharge capacity of PE-BN/PVDF-HFP separator is higher than the Celgard 2325 separator,<sup>[2,14,38]</sup> which indicates the faster transportation of lithium ions. Furthermore, PE-BN/PVDF-HFP separators sustain stable charge-discharge curves in contrast to Celgard 2325 separator, which confirms the enhanced electrochemical reversibility. Figure 5b presents the specific discharge capacity of PE-BN/PVDF-HFP with different weight contents of *h*-BN. The PE-BN/PVDF-HFP separator with 11% of *h*-BN particles owns high discharge capacity as compared to separators with other wt% of *h*-BN. The high discharge capacity of the PE-BN/PVDF-HFP separator with 11 wt% *h*-BN is attributed to the high electrolyte uptake and high ionic conductivity, as shown in Figure S7 (Supporting Information). Figure 5c shows the cycle performance of LFP/Li cells with PE-BN/PVDF-HFP and Celgard 2325 separators at 2 C. LFP/Li cell with PE-BN/PVDF-HFP delivers high discharge capacity of 120 mAh g<sup>-1</sup> after 500 cycles in contrast to Celgard 2325, which delivers 92.9 mAh g<sup>-1</sup> with capacity retention of 95 and 84%, respectively. This enhancement in discharge capacity of the cell with PE-BN/PVDF-HFP separator is attributed to the high electrolyte affinity, interconnected microporous structure which holds the electrolyte for longer time, and high ionic conductivity. Moreover, PE-BN/PVDF-HFP separator yields better C-rate capability in contrast to Celgard 2325 separator from 0.2 to 4 C as shown in Figure 5d. The LFP/Li cell with PE-BN/PVDF-HFP separator delivers 108 mAh g<sup>-1</sup> at 4 C whereas Celgard 2325 shows 78 mAh g<sup>-1</sup>. The cell containing PE-BN/PVDF-HFP represents zero capacity fading even after 60 cycles. The PE-BN/PVDF-HFP bilayer separator exhibits high electrolyte uptake of 348% and thermal shrinkage of 6.6% upon annealing at 140 °C for 1 h, and the LFP/Li cells based on as-prepared bilayer separator deliver stable cycling performances with a capacity retention of 95% at 2 C after 500 cycles at room temperature. The overall performance of the as-prepared separator is enhanced compared with previously reported surface-modified PE separators, as shown in Table S3 (Supporting Information).

The high-temperature cycle performance is evaluated by directly placing the LFP/Li cells in oven at elevated temperatures. Figure 5e shows that the specific discharge capacity of LFP/Li cell based on PE-BN/PVDF-HFP is higher (135 mAh g<sup>-1</sup>) than the cell assembled with Celgard 2325 separator (115 mAh g<sup>-1</sup>) at 60 °C after 80 cycles. The increment in discharge capacity with rise in temperature for LFP/Li cells (with Celgard 2325 and PE-BN/PVDF-HFP separators) owing to the inverse relation of resistance and temperature. After 60 °C, there is a dramatic reduction in discharge capacity observed for both types of separator. The reason behind this reduction



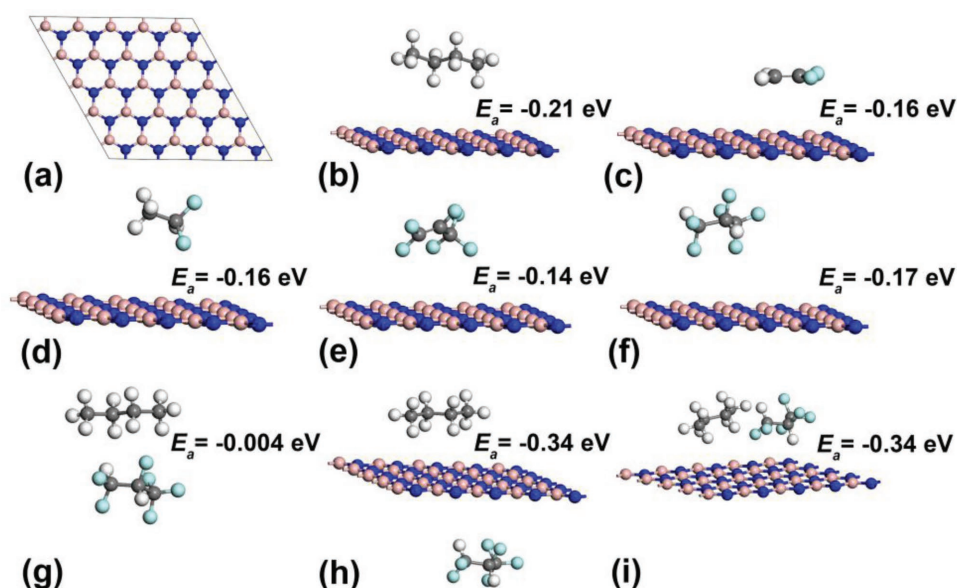


**Figure 5.** a) Charge–discharge voltage–capacity profiles of LFP/Li cells based on Celgard 2325 and PE-BN/PVDF-HFP separators at 2 C, b) specific discharge capacities of LFP/Li cells based on PE-BN/PVDF-HFP with different weight contents of BN at 2 C, c) cycle performance of LFP/Li cells based on PE-BN/PVDF-HFP and Celgard 2325 separators at 2 C after 500 cycles, d) C-rate capabilities of Celgard 2325 and PE-BN/PVDF-HFP separators at room temperature, and e) specific discharge capacity of LFP/Li cells based on Celgard 2325 and PE-BN/PVDF-HFP containing high-temperature electrolyte by direct testing at elevated temperatures (30–70 °C) at 2 C with 40  $\mu\text{m}$  thickness.

in capacity is due to the instability of electrolyte solvents and decomposition of electrolyte salts at  $>60$  °C.<sup>[46–48]</sup> The electrochemical stability of PE-BN/PVDF-HFP and Celgard 2325 separators with lithium anode is evaluated by measuring the symmetric Li/Li cells galvanostatic stability test at a current density of  $0.5\text{ mA cm}^{-2}$  for 400 h at 25 °C. The Li/Li symmetric cell with the PE-BN/PVDF-HFP separator exhibits more stable voltage hysteresis and longer cycling stability (400 h) without any short circuit, as compared to the Li/Li cell with Celgard 2325 separator, which shows severe fluctuation in voltages and becomes short circuited after 126 h, as shown in Figure S8 (Supporting Information). The stable voltages of the Li/bilayer separator/Li symmetric cell confirm that the PE-BN/PVDF-HFP separator mitigates the dendrite growth and penetration,

which avoids internal short circuit. The better stability of the as-prepared separator with lithium anode is attributed to high ionic conductivity, better interfacial adhesion with electrode, superior mechanical strength of incorporated *h*-BN, interconnected microporous structure, and the bilayered formation of the as-prepared separator. These results agree well with the previously reported work of Chen et al.<sup>[49]</sup> and Kang et al.<sup>[50]</sup> They showed that the separator or the coated layer on lithium electrode with the interconnected porous structure can effectively suppress the lithium dendrite growth.

To investigate the effect of *h*-BN particles on interaction between PE-BN and PVDF-HFP layers of the bilayer separator, the interaction energy between PE and PVDF-HFP, with and without incorporation of *h*-BN particles was calculated,



**Figure 6.** Optimized structures of *h*-BN adsorbed with PE and PVDF-HFP molecules and the adsorption energies: a) *h*-BN structure, b) *h*-BN adsorbed with PE, c–f) *h*-BN adsorbed with  $\text{CH}_2\text{CF}_2$ ,  $\text{CH}_3\text{CF}_2$ ,  $\text{CF}_2\text{CFCF}_3$ , and  $\text{CF}_2\text{HCFHCF}_3$  molecules, which stand for different fragments of PVDF-HFP, g)  $\text{CF}_2\text{HCFHCF}_3$  adsorbed with PE, and h,i) *h*-BN adsorbed with PE and  $\text{CF}_2\text{HCFHCF}_3$ .

as shown in **Figure 6**. Four different fluorine contained functional groups including  $\text{CHF}_2\text{CHFCF}_3$ ,  $\text{CH}_2\text{CF}_2$ ,  $\text{CH}_3\text{CHF}_2$ , and  $\text{CF}_2\text{CFCF}_3$  were used as possible active sites of the PVDF-HFP polymer. The calculation results show that *h*-BN owns high adsorption energy with PE and functional groups of PVDF-HFP. Although the PE surface has very weak adsorption interaction with  $\text{CHF}_2\text{CHFCF}_3$  ( $-0.004$  eV), the adsorption interaction of  $\text{CHF}_2\text{CHFCF}_3$  increases to  $-0.34$  eV with the incorporation of *h*-BN between PE and  $\text{CHF}_2\text{CHFCF}_3$  surfaces. The results indicate that the incorporation of *h*-BN particles in PE matrix gives rise to stronger adsorption energy owing to the high interfacial interaction with PVDF-HFP matrix. Thus, the higher affinity of the PE-BN layer with PVDF-HFP layer prevents the separation of PVDF-HFP layer, which further enhances the electrochemical performances of LFP/Li cells with the as-prepared bilayer separator.

### 3. Conclusion

PE-BN/PVDF-HFP bilayer separator is prepared for the first time through wet chemistry method which is more convenient, efficient, and cost-effective method. The incorporation of 2D *h*-BN in PE matrix and the coating of PVDF-HFP gives rise to electrolyte uptake and high thermal stability, and suppresses the formation and growth of dendrites owing to the highly porous interconnected microstructure, the polar nature of *h*-BN and PVDF-HFP, and the high mechanical strength. The as-prepared separator exhibits better thermal stability at  $140^\circ\text{C}$  for 1 h. The LFP/Li cells based on PE-BN/PVDF-HFP bilayer separator deliver cycle performance of  $120\text{ mAh g}^{-1}$  after 500 cycles at  $2^\circ\text{C}$  with 95% capacity retention as well as desirable rate capabilities at room temperature and elevated temperatures. The long cycle performance of LFP/Li cell with PE-BN/PVDF-HFP

resulted from *h*-BN forming the strong interfacial interaction between PE and PVDF-HFP. The above fascinating characteristics with simple and cost-effective preparation method makes PE-BN/PVDF-HFP separator as an effective alternate for traditional multilayer lithium-ion battery separators.

### 4. Experimental Section

**Materials Information:** PVDF-HFP ( $M_w = 100\,000\text{ g mol}^{-1}$ , and 20 wt% HFP) was purchased from Solvay Co. Ltd. (China). PE (particle size =  $25\text{ }\mu\text{m}$ ) was purchased from Zhongshan Lianchang Co. Ltd. (Guangdong, China). Hexagonal-BN powder with particle size  $1\text{ }\mu\text{m}$  and tetrahydrofuran (THF) were purchased from Aladdin Industrial Corporation (Shanghai, China). 1,2-Propanediol and acetone solvents were supplied by Kelong Chemical Reagent Corporation (Chengdu, China). The PE and *h*-BN materials were dried under vacuum at  $70^\circ\text{C}$  for 12 h to eliminate volatile impurities and moisture prior to the preparation of solution. For room-temperature testing, electrolyte solution was obtained from Zhuhai Smoothway Electronic Materials Co., Ltd. (China), comprises of lithium hexafluorophosphate ( $\text{LiPF}_6$ ) salt ( $1\text{ mol L}^{-1}$ ) in a mixture of ethylene carbonate (EC), ethyl methyl carbonate (EMC), diethyl carbonate (DEC), and dimethyl carbonate (DMC) with a composition of (EC/EMC/DMC/DMC = 30/35/15/20 + 2% VC). Whereas, for high-temperature battery testing the electrolyte solution was purchased from Shenzhen Tiancheng Technology Co. Ltd. (China). The trilayer (PP/PE/PP) commercial separator (Celgard 2325) was utilized for comparison. The thickness and areal weight of Celgard 2325 separator were  $25\text{ }\mu\text{m}$  and  $1.72\text{ mg cm}^{-2}$ , respectively.

**Preparation of Separator:** The PE-BN/PVDF-HFP bilayer separator was prepared through a wet chemistry method. For first layer (PE-BN layer), initially, PE powders were added in THF solvent and stirred at  $60^\circ\text{C}$  for 2 h. After dissolution of PE in THF solvent, 1,2-propanediol (as a pore creating agent) and *h*-BN powders were added and stirred for 1 h at  $60^\circ\text{C}$ . The as-prepared solution was coated on iron sheet and dried for 6 h under vacuum at  $30^\circ\text{C}$ . The real-time photos of the complete dissolution process of PE in THF are given in Figure S9 (Supporting Information). The second layer (PVDF-HFP layer) solution



contained PVDF-HFP as a polymer matrix and acetone as a solvent. The solution of PVDF-HFP and acetone was stirred at room temperature for 30 min. After stirring at room temperature, the PVDF-HFP/acetone solution was heated and stirred at 50 °C on a water bath heater for 3–5 min and coated on PE-BN layer as a second layer. The thickness of the PE-BN/PVDF-HFP bilayer separator was maintained to 40 µm with 2.11 mg cm<sup>-2</sup> areal weight.

**Pore-Forming Mechanism in PE-BN Layer:** The PE-BN layer solution was prepared based on two different solvents, THF and 1,2-propanediol. The THF solvent was used for the dissolution of PE, whereas 1,2-propanediol was used as the pore-creating agent. The pore-creating mechanism in the PE-BN layer was based on a bisolvent liberation method. The THF and 1,2-propanediol have boiling points of 66 and 188.2 °C, corresponding to the high difference in evaporation rates. By coating PE-BN solution on iron sheet, the THF solvent evaporates faster at room temperature during solidification and forms porosity. However, 1,2-propanediol appeared in the shape of droplets on the surface of the PE-BN layer and did not evaporate at room temperature, owing to its different evaporation rate compared with THF solvent. 1,2-Propanediol droplets were evaporated during vacuum drying at 30 °C for 6 h and formed dense and interconnected microporous structure. The formed droplets on the surface of PE-BN layer after coating and the SEM images of PE-BN layer with and without 1,2-propanediol are shown in Figure S10 (Supporting Information). The SEM images of PE-BN layer further confirmed that the PE-BN layer with 1,2-propanediol exhibits more porous structure compared with PE-BN layer without 1,2-propanediol. Luo et al.<sup>[9]</sup> also used a bisolvent liberation method (acetone and NMP) for creating interisland and highly porous PVDF-HFP/PE composite separator. Shi et al.<sup>[36]</sup> also investigated the effect of different solvents on the formation of porous structure and concluded that the evaporation rate of different solvents plays a vital role in determining the porous structure.

**Separator Characterization:** The surface morphologies of separators and EDS determinations were observed using a field emission environmental scanning electron microscope (FEI Nova NanoSEM 450). Mercury porosimetry technique was used to analyze the porosity, pore size distribution, and mean pore size of the separators. The porosimetry measurements were conducted by using an Automatic Mercury Porosimeter (PoreMaster-60, Malvern, British) under the pressure ranging from 0.10 to 60 000 psia with a contact angle and a mercury surface tension of 130° and 485.000 dynes cm<sup>-1</sup>, respectively. All samples were dried at 100 °C for 6 h, before porosimetry analysis. The thermal shrinkage was calculated with Equation (S2) (Supporting Information) under heat treatment of separators at different temperatures (50, 75, 100, 125, and 140 °C) for 1 h. The thermal distribution on the surface of separators was examined by using FLIR camera (FLIR-A600-Series, Sweden) by directly placing separators on the heating plate between 30 and 160 °C. The thermal analysis of separator was accomplished by using thermogravimetric analyzer (TGA-103, Nanjing, China) up to 800 °C with a heating rate of 10 °C min<sup>-1</sup> under air atmosphere. The electrolyte retention and electrolyte uptake were obtained by measuring the weight change of separators before and after immersing them in electrolyte solution for different time intervals inside argon (Ar)-filled glove box, as given in Equations (S4) and (S3) (Supporting Information), respectively. The contact angle measurements were conducted with 5 µL droplets of liquid electrolyte on the surface of separators for 20 s by using optical tensiometer (Attension Theta Lite, Sweden) at room temperature. Electromechanical universal testing machine (MTS Systems, CMT6104, China) was used to calculate the mechanical strength of separators.

**Electrochemical Characterization:** The electrochemical performances were carried out on coin cells (LIR2032). The LFP and lithium metal were used as cathode and anode, respectively. The cells were assembled with the configuration of positive coin shell/LFP cathode/electrolyte soaked separator/Li metal anode/current collector/coin cell protector/negative coin shell, inside the glove box (Dellix, Model-LS800S, China) filled with purified Ar gas with water contents <1 ppm. The cycle performance of cells was examined using a battery testing

system (Neware, Model BTS-51, Shenzhen, China) between 2.5 and 4.2 V. Celgard 2325 and PE-BN/PVDF-HFP separators were charged at 2 C for 500 cycles. For galvanostatic cycling measurements, coin cells were assembled with Li metal/electrolyte soaked separator/Li metal configuration inside the glove box (Dellix, Model-LS800S, China) and analyzed using the battery testing system (Neware, Model BTS-51, Shenzhen, China) at 25 °C for 400 h with a current density of 0.5 mA cm<sup>-2</sup>. The OCV, EIS, and CV were observed by assembling cells with the configuration as LFP cathode/electrolyte soaked separator/Li metal by using electrochemical workstation (Model-CHI760E Shanghai, China) at room temperature. The OCV drop measurement was observed with respect to time by placing the charged LFP coin cell in a drying oven at 150 °C for 2500 s. The ionic conductivity ( $\sigma$ ) of the separators was calculated using Equation (S1) (Supporting Information) after measuring the bulk impedance of coin cells configured as steel coin shell/electrolyte soaked separator/steel coin shell by using electrochemical workstation (Model-CHI760E Shanghai, China). The bulk impedance separators were calculated at different temperatures (30–90 °C) in the frequency range of 0.1 Hz to 1 MHz with an amplitude of 5 mV.

**Computational Method:** Density functional theory (DFT) calculations were conducted within the Perdew–Burke–Ernzerhof generalized gradient approximation to calculate adsorption energy values and atomic configurations of *h*-BN with PE and PVDF-HFP by implementing the Dmol3 package. The adsorption energies of PE and functional groups of PVDF-HFP on *h*-BN surface were calculated by subtracting the self-consistent field (SCF) energies of geometry optimized functional groups, PE, and *h*-BN surface from the energy of the optimized PVDF-HFP functional groups and PE adsorbed on the *h*-BN surface. The atomic potentials were evaluated by using double numerical plus polarization (DNP) basis sets with effective core potential. SCF calculations were performed until the SCF tolerance value was less than 1 × 10<sup>-6</sup>.

## Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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## Conflict of Interest

The authors declare no conflict of interest.

## Keywords

bilayer separators, boron nitride, lithium-ion batteries, microporous structure, polyethylene

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